

GAPS IN MULTIPLICATIVE SIDON SETS

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ABSTRACT. For a positive integer n , let $g(n)$ denote the infimum of all real numbers L such that there exists a multiplicative Sidon set $A \subseteq \{1, 2, \dots, n\}$ that intersects every interval $[x, x + L] \subseteq [1, n]$. Sárközy asked for estimates on $g(n)$, and he in particular asked whether one has $g(n) \leq \sqrt{n}$ for every $n \in \mathbb{N}$. We first show that this estimate does indeed hold, with a proof that was autonomously discovered and formally verified in Lean by [Aristotle](#). Next, we improve the upper bound further and, with $\rho = \frac{13 - \sqrt{69}}{10} < 0.47$, prove that $g(n) \ll_{\varepsilon} n^{\rho + \varepsilon}$ for every $\varepsilon > 0$.

1. INTRODUCTION

With $\mathbb{N}$ the set of positive integers, we say that a set $A \subseteq \mathbb{N}$ is a *multiplicative Sidon set* if all products aa' with $a, a' \in A$ and $a \leq a'$ are distinct. Naturally, the set of prime numbers is such a set, from which we deduce that there exist multiplicative Sidon sets in $[n] := \{1, 2, \dots, n\}$ with at least $\pi(n)$ integers. Erdős [4, 5] proved that this is the right asymptotic and determined the correct order of growth of the error term. More precisely, he showed that the maximum size of a multiplicative Sidon set contained in $[n]$ is

$$\pi(n) + \Theta\left(\frac{n^{3/4}}{(\log n)^{3/2}}\right). \quad (1.1)$$

Thus the extremal-size problem is well-understood, even though it is still open what the optimal implied constant is; see e.g. [3]. On the other hand, much less is known about gap sizes in multiplicative Sidon sets. In this regard, Sárközy [8, Problem 34] asked the following questions in his 2001 list of unsolved problems.

Problem 1.1 (Sárközy [8]). *How small can one make the maximal gap between the consecutive elements of a multiplicative Sidon set selected from $\{1, 2, \dots, n\}$? Does there exist for all n a multiplicative Sidon set $A \subseteq \{1, 2, \dots, n\}$ so that $A \cap [m, m + \sqrt{n}] \neq \emptyset$ for all $1 \leq m < m + \sqrt{n} \leq n$?*

As Sárközy [8] points out, if one could prove that the sequence of prime numbers $p_1, p_2, \dots$ satisfies the estimate $p_{i+1} - p_i \leq p_i^{1/2}$, then an affirmative answer to the second question would follow. However, even under the assumption of the Riemann Hypothesis such an estimate is not known.

To quantify the gap size question, let us introduce the following function.

Definition 1.2. For a positive integer n , define $g(n)$ as the infimum of all real numbers $L \geq 0$ for which there exists a multiplicative Sidon set $A \subseteq [n]$ satisfying $A \cap [x, x + L] \neq \emptyset$ for every $x \in \mathbb{R}$ with $[x, x + L] \subseteq [1, n]$.

By combining the prime number theorem, Erdős' estimate (1.1), and the pigeonhole principle, the lower bound $g(n) \geq (1 + o(1)) \log n$ quickly follows.

In the other direction, the best known upper bound on $g(n)$ obtained purely from prime gaps is $g(n) \leq n^{21/40}$ for all large enough n , by a theorem of Baker, Harman and Pintz [2].

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In this paper we manage to lower the exponent and reach the square-root barrier, answering Sárközy's second question affirmatively.

Theorem 1.3. *For all $n \in \mathbb{N}$ one has*

$$g(n) \leq \lfloor \sqrt{n} \rfloor.$$

By making use of two external inputs it is possible to do even better, however. The first is the aforementioned theorem of Baker, Harman and Pintz [2] that every interval $(x - x^{21/40}, x]$ contains a prime for all sufficiently large x . The second is a lower bound of Laishram–Murty for

$$\sum_{1 \leq m \leq x^\beta} \left\{ \pi\left(\frac{x + x^\alpha}{m}\right) - \pi\left(\frac{x}{m}\right) \right\}$$

with suitable parameters α, β [6, Eq. (12)]. This gives the following asymptotic improvement.

Theorem 1.4. *Let*

$$\rho := \frac{13 - \sqrt{69}}{10} \approx 0.46934.$$

Then for every $\varepsilon > 0$ there exists a constant $C_\varepsilon > 0$ such that

$$g(n) \leq C_\varepsilon n^{\rho+\varepsilon}$$

for all $n \in \mathbb{N}$.

The proof of Theorem 1.3 was autonomously discovered and formally verified in Lean by *Aristotle*. The proof of Theorem 1.4 was found by the authors and subsequently formalized in Lean, also by *Aristotle*. Further information on the discovery and verification workflow can be found in the [appendix](#).

2. AN ELEMENTARY CONSTRUCTION

In this section we prove Theorem 1.3.

Proof of Theorem 1.3. We put

$$q := \lfloor \sqrt{n} \rfloor, \quad A := \{a \leq n : a \equiv 1 \pmod{q}\}.$$

It is clear that A intersects every interval of length q , so it is sufficient to show that A is a multiplicative Sidon set. Since $1 + q(q+2) > n$, every element of A has the form $1 + qi$ with $0 \leq i \leq q+1$. Thus it suffices to prove that, whenever i, j, k, ℓ satisfy

$$0 \leq i \leq j \leq q+1 \quad \text{and} \quad 0 \leq k \leq \ell \leq q+1,$$

then the equality

$$(1 + qi)(1 + qj) = (1 + qk)(1 + q\ell)$$

implies $i = k$ and $j = \ell$.

Expanding the brackets on both sides gives

$$1 + q(i + j) + q^2ij = 1 + q(k + \ell) + q^2k\ell.$$

After subtracting 1 and dividing by q , we obtain

$$(i + j - k - \ell) + q(ij - k\ell) = 0. \tag{2.1}$$

With $\Delta := i + j - k - \ell$, we then see $q \mid \Delta$, so either $\Delta = 0$ or $|\Delta| \geq q$.

If $\Delta = 0$, then $ij = k\ell$ by equation (2.1), so the pairs (i, j) and (k, ℓ) have the same sum and the same product. They are therefore the two roots of the same quadratic equation $X^2 - (i+j)X + ij = 0$, and thus $i = k$ and $j = \ell$. It is therefore sufficient to show that the assumption $|\Delta| \geq q$ leads to a contradiction, and by symmetry we may further assume $\Delta \geq q$.

Writing $s := k + \ell$, we see that equation (2.1) implies $s > 0$ and

$$ij < k\ell \leq \left\lfloor \frac{s^2}{4} \right\rfloor. \quad (2.2)$$

From $j \leq q + 1$ and $i + j = s + \Delta$ we get

$$i = s + \Delta - j \geq s - 1.$$

Thus $i \in [s - 1, q + 1]$ and

$$ij = i(s + \Delta - i) \geq i(s + q - i).$$

The latter product is a concave quadratic polynomial in i , so its minimum on $[s - 1, q + 1]$ is attained at an endpoint, which gives $ij \geq (s - 1)(q + 1)$. Now, $\Delta \geq q$ implies

$$s = i + j - \Delta \leq 2q + 2 - q = q + 2.$$

We therefore obtain

$$ij \geq (s - 1)(q + 1) \geq (s - 1)^2 \geq \left\lfloor \frac{s^2}{4} \right\rfloor,$$

contradicting inequality (2.2). This completes the proof. $\square$

3. PRELIMINARIES

3.1. A multiplicative Sidon criterion. The fact that the set of primes is a multiplicative Sidon set can be seen as a special case of the following more general criterion.

Lemma 3.1. *Let $J \geq 1$, and let A be a set of integers such that every $a_i \in A$ can be written in the form $a_i = m_i p_i$, where p_i is a prime, $1 \leq m_i \leq J < p_i$, and the primes p_i are pairwise distinct as a_i ranges over A . Then A is a multiplicative Sidon set.*

Proof. Assume $a_1 a_2 = a_3 a_4$, and suppose by contradiction that $a_1 \notin \{a_3, a_4\}$. Write $a_i = m_i p_i$ with $1 \leq m_i \leq J < p_i$ and p_1 different from p_3 and p_4 . Since $p_1 > J \geq m_3, m_4$, the prime p_1 cannot divide $m_3 m_4$. Hence p_1 must divide $p_3 p_4$, so $p_1 = p_3$ or $p_1 = p_4$, contradicting the assumption that these primes are distinct. Hence, $a_1 \in \{a_3, a_4\}$ and cancelling the common factor gives $\{a_1, a_2\} = \{a_3, a_4\}$. $\square$

3.2. Primes in short intervals. As we alluded to before, we shall use the following result of Baker, Harman and Pintz [2, Theorem 1].

Lemma 3.2 ([2]). *For all sufficiently large real numbers x , the interval $(x - x^{21/40}, x]$ contains a prime.*

We will also use the following consequence of [6, Eq. (12)].

Lemma 3.3. *Let $0 < \beta < \alpha < \frac{1}{2}$, and let $\delta > 0$ satisfy $3\alpha - 4/3 < \delta < (5\alpha - 2)/3$. Set*

$$\eta := 1 - \alpha - \frac{1 - \alpha^2 - \beta(2 - \beta)}{\delta}.$$

Then for every real number $c_0 < \eta$, one has

$$\sum_{1 \leq m \leq x^\beta} \left(\pi\left(\frac{x + x^\alpha}{m}\right) - \pi\left(\frac{x}{m}\right) \right) \geq c_0 x^\alpha \quad (3.1)$$

for all sufficiently large x . Moreover, the primes counted on the left-hand side of (3.1) are distinct. Consequently, for all sufficiently large x , there exist at least $c_0 x^\alpha$ distinct primes p for which $mp \in (x, x + x^\alpha]$ for some integer $m \leq x^\beta$.

Proof. Since $0 < \beta < \alpha < \frac{1}{2}$ and $3\alpha - 4/3 < \delta < (5\alpha - 2)/3$, the estimate stated in [6, Eq. (12)] applies with our parameters α, β, δ . More precisely, for every $\varepsilon' > 0$ and all sufficiently large x ,

$$\sum_{1 \leq m \leq x^\beta} \left(\pi\left(\frac{x + x^\alpha}{m}\right) - \pi\left(\frac{x}{m}\right) \right) \geq \left(1 - \alpha - \varepsilon' - \frac{1 - \alpha^2 - \beta(2 - \beta)}{\delta} \right) x^\alpha.$$

Since $c_0 < \eta$, we may choose $\varepsilon' > 0$ so small that

$$1 - \alpha - \varepsilon' - \frac{1 - \alpha^2 - \beta(2 - \beta)}{\delta} > c_0,$$

proving (3.1). It remains to prove the distinctness statement, which is essentially equivalent to the inequality $\frac{x+x^\alpha}{m+1} \leq \frac{x}{m}$ for all $1 \leq m \leq x^\beta$. Multiplying this inequality by $m(m+1)$ gives $mx^\alpha \leq x$, which holds for all $m \leq x^\beta$ since $\alpha + \beta < 1$. The final assertion follows immediately. $\square$

3.3. A weighted Hall-type lemma. We shall also use the following weighted variant of Hall's theorem.

Lemma 3.4. *Let $G = (\mathcal{L}, \mathcal{R}, \mathcal{E})$ be a finite bipartite graph, and suppose that each edge $e \in \mathcal{E}$ is assigned a non-negative weight $w(e)$. Assume that there exists a real number $L_0 > 0$ such that*

$$\sum_{\substack{e \in \mathcal{E} \\ e \ni u}} w(e) \geq L_0 \quad \text{for every } u \in \mathcal{L},$$

and

$$\sum_{\substack{e \in \mathcal{E} \\ e \ni v}} w(e) \leq L_0 \quad \text{for every } v \in \mathcal{R}.$$

Then G has a matching covering all vertices of $\mathcal{L}$.

Proof. Let $S \subseteq \mathcal{L}$, and let $N(S) \subseteq \mathcal{R}$ be its neighborhood. Then

$$L_0|S| \leq \sum_{u \in S} \sum_{\substack{e \in \mathcal{E} \\ e \ni u}} w(e).$$

Every edge counted on the right-hand side has its $\mathcal{L}$ -endpoint in S , hence its $\mathcal{R}$ -endpoint lies in $N(S)$. Therefore

$$\sum_{u \in S} \sum_{\substack{e \in \mathcal{E} \\ e \ni u}} w(e) \leq \sum_{v \in N(S)} \sum_{\substack{e \in \mathcal{E} \\ e \ni v}} w(e) \leq L_0|N(S)|.$$

Thus $|S| \leq |N(S)|$. So Hall's condition holds, and Hall's theorem yields a matching covering $\mathcal{L}$. $\square$

4. PROOF OF THE POWER-SAVING BOUND

Proof of Theorem 1.4. Since $\rho = \frac{13-\sqrt{69}}{10} < \frac{19}{40}$, without loss of generality we may assume that $0 < \varepsilon < \frac{19}{40} - \rho$. Choose α so that $\rho < \alpha < \rho + \varepsilon$. Then $\alpha < \frac{19}{40}$ and $5\alpha^2 - 13\alpha + 5 < 0$, where the last inequality is equivalent to

$$1 - \alpha - \frac{1 - 2\alpha}{(5\alpha - 2)/3} > 0.$$

We now define

$$F(\beta, \delta) := 1 - \alpha - \frac{1 - \alpha^2 - \beta(2 - \beta)}{\delta}.$$

Then $F(\alpha, (5\alpha - 2)/3) > 0$, while $\alpha < \rho + \varepsilon < 1/2$ implies $3\alpha - 4/3 < (5\alpha - 2)/3$. By continuity of F , we may therefore choose real numbers β, δ such that

$$0 < \beta < \alpha < \frac{1}{2}, \quad 3\alpha - \frac{4}{3} < \delta < \frac{5\alpha - 2}{3},$$

and $\eta := F(\beta, \delta) > 0$. Applying Lemma 3.3 with $c_0 := \eta/2 > 0$ then gives

$$\sum_{1 \leq m \leq x^\beta} \left(\pi\left(\frac{x + x^\alpha}{m}\right) - \pi\left(\frac{x}{m}\right) \right) \geq c_0 x^\alpha \quad (4.1)$$

for all sufficiently large x .

Now let n be sufficiently large and define

$$H := \lceil 2n^\alpha \rceil, \quad J := \lfloor n^\beta \rfloor, \quad T := \lfloor n/H \rfloor, \quad t := \lfloor H^{19/21} \rfloor.$$

With these definitions we further define the intervals

$$B_i := (iH, (i+1)H] \quad (1 \leq i < T).$$

By Lemma 3.1 it is sufficient to find integers $m_1, \dots, m_{T-1} \leq J$ and distinct primes $p_1, \dots, p_{T-1} > J$ such that $m_i p_i \in B_i$ for all $1 \leq i < T$. Indeed, the first element $m_1 p_1 \in B_1$ is then at most $2H$, the difference between two consecutive elements is smaller than $2H$, and the largest element $m_{T-1} p_{T-1} \in B_{T-1}$ is larger than $n - 2H$. Assuming such m_i and p_i are chosen, with $C_\varepsilon = 5$ we would then get $g(n) \leq 2H \leq C_\varepsilon n^{\rho+\varepsilon}$, and we can cover the finitely many remaining values of n by possibly enlarging C_ε .

Now, for $i < t$, the largest element of B_i is $(i+1)H$, while the length of B_i is

$$H = H^{19/40} H^{21/40} \geq (tH)^{21/40} \geq ((i+1)H)^{21/40}.$$

Lemma 3.2 therefore implies that for every $i < t$, B_i contains a prime p_i . Moreover, for all such i we have

$$p_i \geq p_1 > H > J.$$

We therefore choose $m_i = 1$ for all $1 \leq i < t$.

As for the intervals B_i with $t \leq i < T$, with $x_i := iH$ the left endpoint of B_i , define

$$\mathcal{R}_i := \{p : p \notin \{p_1, \dots, p_{t-1}\} \text{ is prime and } mp \in B_i \text{ for some positive integer } m \leq x_i^\beta\}.$$

For $p \in \mathcal{R}_i$ we note that we also have

$$p > \frac{x_i}{x_i^\beta} \geq x_t^{1-\beta} = (tH)^{1-\beta} > (n^{40\alpha/21})^{21/40} = n^\alpha > J.$$

We now consider the bipartite graph with left vertex set $\mathcal{L} := \{B_t, \dots, B_{T-1}\}$ and right vertex set $\mathcal{R} := \bigcup_{i=t}^{T-1} \mathcal{R}_i$. We then join a vertex $B_i \in \mathcal{L}$ to a prime $p \in \mathcal{R}_i$, and we define the weight of such an edge (B_i, p) by

$$w(B_i, p) := \sum_{\substack{1 \leq m \leq x_i^\beta \\ mp \in B_i}} \frac{1}{m}.$$

As the intervals B_i are disjoint, every fixed m occurs in the summation on the right-hand side for at most one index i . Hence, for all $p \in \mathcal{R}$ we have

$$\sum_{B_i \in \mathcal{L}} w(B_i, p) \leq \sum_{1 \leq m \leq n} \frac{1}{m}.$$

We therefore deduce by Lemma 3.4 that it now suffices to show that for all $t \leq i < T$ we have

$$\sum_{p \in \mathcal{R}} w(B_i, p) \geq \sum_{1 \leq m \leq n} \frac{1}{m}. \quad (4.2)$$

Indeed, this would imply the existence of a matching, which gives us for all $t \leq i < T$ a prime $p_i > J$ such that $m_i p_i \in B_i$ for some $m_i \leq \lfloor x_i^\beta \rfloor \leq \lfloor n^\beta \rfloor = J$.

In order to show (4.2), fix i and m . There are at most two primes $p \in \{p_1, \dots, p_{t-1}\}$ such that $mp \in B_i$. Indeed, if p_j, p_k, p_ℓ were three such primes with $1 \leq j < k < \ell < t$, then $\ell \geq j + 2$, and since $p_j \in B_j$ and $p_\ell \in B_\ell$ we have $p_\ell - p_j > H$. Thus

$$mp_\ell - mp_j = m(p_\ell - p_j) > mH \geq H,$$

contradicting the fact that both mp_j and mp_ℓ lie in the interval B_i , which has length H . Secondly, as $x_i^\alpha < n^\alpha < H$, the interval $(x_i, x_i + x_i^\alpha]$ is contained in B_i . Hence, applying equation (4.1) gives

$$\begin{aligned} \sum_{p \in \mathcal{R}} w(B_i, p) &\geq \sum_{1 \leq m \leq x_i^\beta} \frac{1}{m} \left(\pi\left(\frac{x_i + x_i^\alpha}{m}\right) - \pi\left(\frac{x_i}{m}\right) \right) - \sum_{\substack{1 \leq m \leq x_i^\beta \\ mp \in B_i \\ p \in \{p_1, \dots, p_{t-1}\}}} \frac{1}{m} \\ &\geq c_0 x_i^{\alpha-\beta} - 2 \sum_{1 \leq m \leq n} \frac{1}{m} \\ &\geq c_0 n^{\alpha(\alpha-\beta)} - 2 \sum_{1 \leq m \leq n} \frac{1}{m} \\ &> 3 \log(3n) - 2 \log(3n) \\ &> \sum_{1 \leq m \leq n} \frac{1}{m}. \end{aligned}$$

□

APPENDIX: FORMAL DISCOVERY AND VERIFICATION WITH ARISTOTLE

The proof of Theorem 1.3 was autonomously discovered and formally verified in Lean by *Aristotle*, a formal reasoning agent developed by Harmonic [1] and publicly available for free at aristotle.harmonic.fun.

We first asked Aristotle to investigate the second subquestion in Problem 1.1. It autonomously produced a proof of Theorem 1.3, together with a Lean formalization. After that, the human authors studied the first subquestion in Problem 1.1 more carefully and managed to prove the stronger conclusion stated in Theorem 1.4. We then asked Aristotle to formalize the latter proof, under the assumption of the results by Baker–Harman–Pintz and Laishram–Murty that we used. Aristotle successfully completed this second (conditional) formalization as well.

The full Lean file combining both formalizations can be inspected interactively in the [Lean 4 Web Editor](#).

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